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Roll No

EC-6002-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any missing data.

1. a) State whether the following systems are (i) Static (ii) Linear (iii) Stiff invariant (iv) Causal and (v) Stable.

i) $y(n) = n x(n)$

ii) $y(n) = x(n^2)$

iii) $y(n) = x(n) + 3u(n+1)$

iv) $y(n) = g(n)x(n)$. Here $g(n)$ is another sequence.

b) Discuss causal and non-causal LTI systems. Give examples of each system. What is the significance of causality of LTI systems? Explain.

2. a) A discrete - time system is characterized by the following difference equation:

$$y(n) = x(n) + e^{\alpha} y(n-1)$$

Check this system for BIBO stability.

b) Find the discrete-time Fourier transform of unit-step sequence $x(n) = u(n)$.

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3. a) Find z-transform of the following discrete-time signal
$$x(n) = 2^n u(n-2)$$

b) State and prove the convolution property of z-transform.
4. a) Find the inverse z-transform of:
$$X(z) = \frac{3}{z-2} \quad |z| > 2$$

b) Explain Discrete Fourier Series. What is the relationship between DFT and DFS. Write down the properties of DFT.
5. a) Compute circular convolution of the following two sequences:
$$x_1(n) = \left\{ \underset{\uparrow}{1}, 2, 0, 1 \right\}$$
$$x_2(n) = \left\{ \underset{\uparrow}{2}, 2, 1, 1 \right\}$$

b) Explain in detail how DFT can be used as a linear transformation tool in Digital Signal Processing.
6. a) Obtain DFT of the following sequence:
$$x(n) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0 \right)$$

Using decimation in frequency FFT algorithm.
b) Discuss the FFT algorithms for composite value of N.
7. a) Differentiate between FIR filters and IIR filters. Write the design techniques for designing FIR filters.
b) What do you mean by bilinear transformation? Write down its advantages and disadvantages.

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8. Write short notes on (any two):
- i) Impulse Invariance Method
 - ii) Examples of FIR filters
 - iii) Windowing Techniques
 - iv) Signal flow graphs
